

MODULE 1 – DETAILED REVIEWER

Foundations of Business and IS Performance Evaluation

1. Business Performance Evaluation

Definition (Thorough Explanation)

Business Performance Evaluation is a **hybrid field combining strategic management, organizational behavior, and management accounting** that systematically assesses how effectively an organization achieves its objectives.

- From **strategic management**, it ensures that organizational actions align with long-term goals and competitive positioning.
- From **organizational behavior**, it considers how people, culture, leadership, and structure influence performance.
- From **management accounting**, it uses financial and non-financial data to measure results and efficiency.

It is described as a “**mechanism for organizational control**”, meaning it provides a structured way to:

- monitor operations
- compare actual performance against planned objectives
- implement corrective actions when deviations occur

In a deeper sense, it serves both:

- a **diagnostic function** (identifying problems and inefficiencies), and

- a **developmental function** (guiding improvements and strategic adjustments).
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Purpose (Thorough Explanation)

Business Performance Evaluation aims to:

- assess the **overall health and competitiveness** of the organization
- identify **performance gaps** between current and desired outcomes
- support **evidence-based decision-making**
- provide **insights for growth, efficiency, and innovation**

It integrates both:

- **financial indicators** (e.g., revenue, profit, cost efficiency), and
 - **non-financial indicators** (e.g., customer satisfaction, operational efficiency, innovation capability).
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2. Business Performance vs. IS Performance

Business Performance (Expanded Definition)

Business Performance refers to the **comprehensive efficacy of a business in fulfilling its strategic goals across various dimensions such as profitability, market share, and how well it keeps customers.**

- It reflects the **overall success of the organization**
- It evaluates whether strategic objectives are achieved

- It includes both **short-term results** and **long-term sustainability**

In deeper terms, Business Performance is **multi-dimensional**, meaning it cannot be measured by a single metric but requires a combination of indicators.

IS Performance (Expanded Definition)

IS Performance pertains to the **effectiveness and efficiency of the systems and focuses on the underlying factors that make a system work, such as system uptime, data accuracy, and transaction speed.**

- It evaluates how well **Information Systems support operations**
- It focuses on **technical and operational efficiency**
- It measures system reliability, responsiveness, and accuracy

In deeper terms, IS Performance ensures that:

- systems deliver **timely and accurate information**
 - processes are **automated and efficient**
 - users experience **minimal delays and errors**
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Relationship Between the Two

- High **IS Performance** contributes to improved **Business Performance**, but it is **not sufficient on its own.**
- Business Performance depends on **strategy, people, and processes**, not just technology.

This means:

- Efficient systems enable better performance
 - But poor strategy or management can still lead to weak business outcomes
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3. Management to Leadership

Traditional Performance Management

The module states that traditional views of performance management are often associated with:

- bureaucracy
- rigid structures
- annual performance appraisals

These approaches are considered **obsolete** because they:

- focus on periodic evaluation rather than continuous improvement
 - emphasize control rather than development
 - react to problems instead of preventing them
-

Performance Leadership (Expanded Definition)

Performance leadership is a modern approach that focuses on:

- setting clear goals
- fostering accountability
- optimizing efficiency
- enabling continuous improvement

In deeper terms, it shifts from:

- evaluating employees → to **developing and empowering them**
- controlling processes → to **guiding and improving them**

It emphasizes:

- proactive decision-making
- coaching and mentoring
- alignment of individual performance with organizational goals

Modern Evaluation Concept

Evaluation is now described as a **continuous process of identifying, measuring, and developing talent through high-frequency feedback**.

This means:

- evaluation happens regularly, not just annually
- feedback is immediate and actionable
- focus is on **outcomes and impact**, not just activity

4. Strategic Visualization: The Balanced Scorecard

Definition (Expanded)

The **Balanced Scorecard** is a framework used to translate strategy into measurable performance indicators across multiple perspectives.

It ensures that organizations evaluate performance **holistically**, rather than relying solely on financial metrics.

Four Perspectives (Thorough Explanation)

1. Financial

Focuses on the efficient use of financial resources to create shareholder value.

It answers whether the organization is achieving profitability and financial sustainability.

2. Customers

Examines how the organization is perceived by customers and how well it delivers value.

It focuses on satisfaction, retention, and loyalty.

3. Internal Processes

Evaluates the efficiency and effectiveness of internal operations that deliver products or services.

It identifies bottlenecks and areas for process improvement.

4. Learning and Growth (Organizational Capacity)

Focuses on the infrastructure required for long-term improvement, including people, systems, and organizational culture.

It ensures the organization can innovate and adapt over time.

Integrated Perspective

These four perspectives are interconnected through **cause-and-effect relationships**:

- Improvements in learning and growth enhance processes
- Better processes improve customer satisfaction
- Satisfied customers lead to improved financial results

5. Quality Management and Continuous Improvement

Definition (Expanded)

Quality Management refers to a set of fundamental beliefs and practices used to guide an organization toward improved performance and sustained success.

It is closely linked to **continuous improvement**, which is described as a **pervasive philosophy** embedded in all organizational activities.

The Seven Principles of Quality Management (Expanded)

1. **Customer Focus**
Organizations depend on customers and must meet or exceed their expectations.
This principle ensures long-term success through customer satisfaction and loyalty.
2. **Leadership**
Leaders establish unity of purpose and direction.
They create conditions where people are engaged in achieving objectives.
3. **Engagement of People**
Competent and empowered employees contribute to organizational success.
Engagement leads to innovation, accountability, and improved performance.
4. **Process Approach**
Activities are managed as interconnected processes, improving consistency and efficiency.

5. Improvement

Ongoing improvement is essential for maintaining competitiveness and adaptability.

6. Evidence-Based Decision Making

Decisions are based on the analysis of data and information, ensuring objectivity and accuracy.

7. Relationship Management

Organizations manage relationships with stakeholders to optimize performance and create value.

6. Continuous Improvement Methodologies

PDCA Cycle

A **four-step iterative method** used to control and improve processes:

- Plan: Identify objectives and processes
- Do: Implement the plan
- Check: Monitor and evaluate results
- Act: Take corrective actions

Kaizen

A philosophy that emphasizes **small, incremental changes** made regularly by all employees.

Over time, these small improvements result in significant overall gains.

Six Sigma

A **data-driven methodology** that uses statistical analysis to reduce process variation and eliminate defects.
Its goal is to achieve near-perfect quality.

7. Problem Identification and Analysis Tools

5 Whys

A technique that involves repeatedly asking “why” to uncover the **root cause** of a problem.

Fishbone Diagram (Cause-and-Effect Diagram)

A visual tool used to identify and categorize possible causes of a problem into areas such as:

- people
- processes
- equipment
- materials
- environment

Pareto Analysis/Chart

Based on the **80/20 rule**, indicating that a small number of causes are responsible for the majority of problems. It helps prioritize issues with the greatest impact.

Process Mapping/Flowcharts

Visual representations of processes that help identify:

- inefficiencies
 - redundancies
 - bottlenecks
-

Value Stream Mapping (VSM)

A detailed method of process mapping that highlights both:

- value-adding activities
 - non-value-adding activities (waste)
-

8. Implementation and Control Tools

Gemba Walks

Managers observe actual work processes in the workplace to gain real insights and identify improvement opportunities.

Kanban

A visual workflow system that manages tasks and limits work-in-progress to ensure efficiency and prevent bottlenecks.

5S System

A workplace organization method consisting of:

- Sort
 - Set in order
 - Shine
 - Standardize
 - Sustain
-

Control Charts

Statistical tools used to monitor process stability over time and detect variations.

Poka-Yoke (Mistake-Proofing)

Techniques designed to prevent errors before they occur by making mistakes difficult or impossible.

Standardized Work

Documented procedures that ensure consistency, efficiency, and a baseline for improvement.

9. Other Relevant Methodologies

DMAIC

A structured problem-solving approach consisting of:

- Define
 - Measure
 - Analyze
 - Improve
 - Control
-

A3 Problem Solving

A structured and concise method of documenting and solving problems, typically presented on a single page.

Just-in-Time (JIT)

A production strategy that delivers products or materials only when needed, reducing waste and improving efficiency.

MODULE 2 – DETAILED REVIEWER

Strategic Metrics & Measurement (KPIs, KRAs, & Quality)

1. Key Result Areas (KRAs)

Definition (Thorough Explanation)

Key Result Areas (KRAs) are the **broad, strategic areas of responsibility or objectives within an organization or job role where individuals or teams must achieve favorable outcomes to succeed in their overall mission.**

- KRAs define **what must be accomplished**, not how it is done
- They represent **critical outcomes**, not daily activities
- They are aligned with **organizational goals and strategy**

In deeper terms, KRAs:

- translate high-level strategy into **focus areas**
 - ensure that each role contributes to **organizational success**
 - act as the foundation for defining measurable indicators (KPIs)
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Characteristics of KRAs (Expanded)

1. Outcome Oriented

KRAs describe results, not activities. Example: “Financial reporting” instead of “prepare reports”

2. Controllable

The person or department must have authority over the outcome.

3. Essential

KRAs represent critical areas where failure would significantly affect the organization.

Identifying KRAs (Step-by-Step Explanation from Pages 3–4)

1. **Understand the Strategic Context**
Align KRAs with mission, vision, and strategic goals.
 2. **List Main Activities and Responsibilities**
Identify all tasks, then focus on the most impactful ones using Pareto principle (80/20 rule).
 3. **Translate Activities into Result Areas**
Convert actions into outcomes (remove verbs and focus on results).
 4. **Collaborate and Gain Alignment**
Ensure agreement among supervisors, employees, and stakeholders.
 5. **Document and Define KPIs**
Clearly define KRAs and pair them with measurable KPIs using SMART criteria.
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2. Key Performance Indicators (KPIs)

Definition (Thorough Explanation)

Key Performance Indicators (KPIs) are specific, quantifiable, and time-bound metrics used to track and evaluate progress toward achieving the broader objectives defined by KRAs.

- KPIs measure **how well objectives are achieved**
 - They are derived from KRAs and strategy
 - They serve as **critical success factors**
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Characteristics of KPIs (Expanded)

- **Specific** – clearly defined and unambiguous
- **Measurable** – expressed using objective data
- **Achievable** – realistic given resources and constraints
- **Relevant** – aligned with business objectives
- **Time-bound** – includes a defined timeframe

These follow the **SMART framework**.

Core Purposes of KPIs (Page 5 Table)

1. **Measure Progress**
Tracks performance over time toward goals
 2. **Guide Decision-Making**
Provides data for informed operational and strategic decisions
 3. **Promote Accountability and Alignment**
Ensures employees understand their contribution
 4. **Benchmark Performance**
Enables comparison with past results or industry standards
-

Types of KPIs (Page 5 Table)

1. **Financial**
Measures profitability and financial health
Example: net profit margin, ROI
2. **Customer**
Measures satisfaction and retention

Example: Customer Lifetime Value (CLV), Net Promoter Score (NPS)

3. **Operational/Process**

Measures efficiency and productivity
Example: order fulfillment time, production efficiency

4. **Human Resources**

Measures employee performance and retention
Example: turnover rate, engagement

5. **Sales & Marketing**

Measures effectiveness of campaigns
Example: conversion rate, lead-to-customer ratio

Lagging vs Leading Indicators (Page 6)

- **Lagging Indicators**
Measure past performance (e.g., profit, revenue)
- **Leading Indicators**
Predict future performance (e.g., number of leads, pipeline growth)

A balanced system uses both for **comprehensive evaluation**.

3. Metrics and Strategic Alignment

Definition of Metrics

A **Metric** is a **standard of measurement used to quantify the performance of processes, activities, or outcomes**.

- Metrics provide **data points**
- They become KPIs when aligned with strategic objectives

Characteristics of Metrics

- Can be qualitative or quantitative, but performance evaluation emphasizes **quantifiable measures**
- Must be **linked to objectives** to have value
- Otherwise, they remain isolated data points

Importance of Aligning Metrics to Business Objectives (Pages 7–8)

1. **Ensures Strategic Focus and Prioritization**
Resources are directed toward high-impact areas
2. **Drives Effective, Data-Driven Decision Making**
Prevents reliance on intuition
3. **Fosters Accountability and Clarity**
Employees understand how their work contributes
4. **Prevents “Vanity Metrics”**
Avoids focusing on irrelevant metrics that do not drive outcomes

4. The Balanced Scorecard (BSC) and KPIs
Definition

The **Balanced Scorecard (BSC)** is a **primary management tool used to force alignment across a balanced set of measures**.

Four Perspectives with KPI Alignment (Page 8 Table)

1. **Financial**
Focus: profitability and shareholder value

Example KPI: Return on Investment (ROI)

2. **Customer**

Focus: satisfaction and retention

Example KPI: Net Promoter Score (NPS)

3. **Internal Processes**

Focus: efficiency and quality

Example KPI: cycle time, defect rate

4. **Learning & Growth**

Focus: innovation and human capital

Example KPI: % of staff trained

Key Insight

The BSC transforms strategy into:

- measurable objectives
- aligned KPIs
- coordinated organizational effort

5. Measuring Customer Satisfaction for Business and IS

Definition (Page 9)

Quality metrics are used to evaluate:

- performance
- efficiency
- effectiveness of a process or product

However:

- business managers and IT managers define “quality” differently

General Steps for Developing Quality Metrics

1. Involve stakeholders

2. Define goals
3. Establish baselines
4. Align with SMART
5. Balance indicators
6. Implement and communicate
7. Monitor and review

6. Business vs IS Metrics (Page 10 Table)

Business Metrics (Outcomes)

- Cost per unit
- Customer churn rate
- Time-to-market speed
- Return rate

IS/IT Metrics (Enablers)

- Data throughput
- System uptime
- Latency
- Defect density

Key Idea:

- Business metrics measure **results**
- IS metrics measure **system performance that enables results**

7. Key Quality Metrics and Formulas

1. Defect Rate / Defect Density

Measures the percentage of defects in a product or system

Formula:

$(\text{Number of Defects} / \text{Total Units}) \times 100$

2. First Pass Yield (FPY)

Measures the percentage of outputs completed correctly the first time

Formula:

$$\left(\frac{\text{Number of Issues Resolved on First Contact}}{\text{Total Contacts}} \right) \times 100$$

3. Customer Satisfaction Score (CSAT)

Measures customer satisfaction using ratings

Formula:

$$\left(\frac{\text{Number of Satisfied Responses}}{\text{Total Responses}} \right) \times 100$$

4. Net Promoter Score (NPS)

Measures customer loyalty

Formula:

$$\% \text{ Promoters} - \% \text{ Detractors}$$

5. On-Time Delivery / SLA Compliance

Measures percentage of services delivered as promised

6. Cost of Poor Quality (COPQ)

Measures financial impact of defects and inefficiencies

8. Measuring and Analyzing Customer Satisfaction

Types of Metrics (Page 13 Table)

1. **CSAT** – measures immediate satisfaction
2. **CES (Customer Effort Score)** – measures ease of interaction

3. **NPS** – measures long-term loyalty
-

Advanced Metrics

- **Customer Health Score**
- **Net Emotional Value (NEV)**

These go beyond satisfaction to analyze:

- emotions
 - loyalty risk
 - churn likelihood
-

9. SERVQUAL Model (Page 14)

Definition

A diagnostic tool used to identify gaps between:

- customer expectations
 - actual service performance
-

Five Dimensions (RATER)

1. **Reliability** – ability to perform service dependably
 2. **Assurance** – knowledge and courtesy of employees
 3. **Tangibles** – physical facilities and equipment
 4. **Empathy** – individualized attention
 5. **Responsiveness** – willingness to help customers
-

10. Predictive Analytics and Customer Retention

Churn Rate

Measures percentage of customers lost

Formula:

$(\text{Customers Lost} / \text{Total Customers}) \times 100$

RFM Model

- Recency
- Frequency
- Monetary Value

Used to predict future customer behavior

ACSI Model

Links customer satisfaction to:

- expectations
 - perceived quality
 - perceived value
-

11. Voice of Customer (VoC) Program

Definition

An operational system that collects and analyzes customer feedback.

Steps

1. **Listen (Intake)**
Surveys, feedback, behavioral data
2. **Analyze (AI Revolution)**
NLP and sentiment analysis
3. **Act (Closed Loop)**
Take action and follow up with customers

MODULE 3 – DETAILED REVIEWER

Technology Enablers: Business Intelligence & Data Warehousing

1. Data Warehouse as Phase 1 of Business Performance Evaluation

Definition (Thorough Explanation)

A **Data Warehouse** is introduced as the **foundation for evaluating business performance**, where organizations first build a **structured repository of integrated data** before applying analytics.

According to the discussion (page 2), business performance evaluation must follow a structured journey:

1. **Build reliable data (Data Warehousing)**
 2. **Apply analytical lens (Business Intelligence)**
-

The Problem: “Single Source of Truth”

In most organizations, data exists in **silos**:

- Sales → CRM
- Finance → ERP
- Marketing → spreadsheets

This leads to:

1. **Inconsistency**
Example: Sales says revenue is 1.2M, Finance says 1.0M
 2. **System Overload**
Running analytics directly on operational systems slows or crashes them
-

Phase 1 Solution

The Data Warehouse:

- extracts data from multiple sources
- integrates and standardizes it
- creates a “**single version of truth**”

This ensures:

- consistent metrics
- unified definitions (e.g., “Net Profit”)
- reliable reporting

2. Data Consolidation and Integration

Definition (Page 3 Explanation)

The **primary task of a data warehouse** is to extract data from every operational source (ERP, CRM, finance, etc.) and consolidate it into one system.

Before Data Warehouse

- Data is scattered across departments
- Metrics are inconsistent
- Manual reconciliation is required

After Data Warehouse

- Data is **standardized and integrated**
- Everyone uses the same definitions
- Reports are consistent across departments

3. Data Cleansing and Quality Assurance

Definition

This phase ensures that data is:

- cleaned
- validated
- standardized

Key Concept: “Garbage In, Garbage Out”

If poor-quality data enters the system:

- all analysis will be inaccurate

Therefore:

- data governance
- early-stage cleansing are critical

4. Historical Data Capture

Definition (Page 4)

A Data Warehouse is designed to be:

- **time-variant** (stores data over time)
- **non-volatile** (data is not overwritten)

Importance

- Enables trend analysis
- Supports performance evaluation over time

Example:

- Compare quarterly sales trends
- Analyze long-term customer behavior

5. Structuring Data for Analysis (OLAP Foundation)

Definition

Data is structured using:

- **Star schema**
- **Snowflake schema**

to optimize performance for reporting and analytics.

Key Insight

Operational databases (OLTP) are optimized for transactions

Data warehouses are optimized for **analysis and reporting (OLAP)**

6. Business Intelligence (BI)

Definition (Page 4)

Business Intelligence (BI) is a **technology-driven process for analyzing data and presenting actionable information to help executives and decision-makers make informed decisions.**

BI as a Process

BI transforms:

- raw data → meaningful insights

It involves:

- data collection
 - transformation
 - analysis
 - visualization
-

7. Business Intelligence Architecture

Components (Page 5)

1. **Data Sources**
CRM, ERP, databases, spreadsheets

2. **ETL Process (Extract, Transform, Load)**

- Extract data from sources
- Transform into usable format
- Load into warehouse

3. **Data Warehouse**

Central storage for structured data

4. **BI Tools (Analytics & Reporting)**

Dashboards, reports, visualizations

Key Insight

The architecture ensures that:

- data flows systematically
 - analysis is efficient and reliable
-

8. OLAP: Online Analytical Processing

Definition

OLAP allows users to analyze data from multiple perspectives using multidimensional structures.

Key Concept

Data is treated like a **“cube”**, where each dimension represents a business perspective.

Dimensions of OLAP (Page 5)

1. **Time**
Year → Quarter → Month → Day
2. **Location/Geography**
Country → Region → City

3. **Product**
Category → Brand → Product
 4. **Customer**
Segment → Individual
 5. **Salesperson/Employee**
-

Capability

Users can:

- drill down (more detail)
 - roll up (summary view)
-

9. BI Dashboards

Definition (Page 6)

A **BI Dashboard** is a visual interface that presents key performance indicators in an easy-to-understand format.

Purpose

- Provide **at-a-glance insights**
 - Support real-time decision-making
 - Track business performance
-

10. Four Levels of Business Intelligence Analytics

1. Descriptive Analytics

- Focus: What happened?
 - Uses reports and dashboards
-

2. Diagnostic Analytics

- Focus: Why did it happen?

- Uses data exploration and root cause analysis
-

3. Predictive Analytics

- Focus: What will happen?
 - Uses statistical models and patterns
-

4. Prescriptive Analytics

- Focus: What should we do?
 - Suggests actions based on predictions
-

Key Insight

These levels represent **maturity in data-driven decision making**.

11. Common BI Tools (Page 6–7)

- **Microsoft Power BI**
Known for integration with Microsoft ecosystem
 - **Tableau**
Known for data visualization and dashboards
 - **Qlik Sense**
Known for associative data exploration
-

Factors in Choosing BI Tools

- business needs
- budget
- technical infrastructure
- user skill level

12. Role and Structure of Data Warehouse

Key Functions (Page 8)

1. **Storing History**
Maintains past data for analysis
2. **Improving Key Targets**
Enables KPI tracking and variance analysis

13. Dimensional Modeling

Star Schema (Page 8)

- Simple structure
- Central fact table + dimension tables
- Faster queries

Snowflake Schema (Page 9)

- More normalized
- More complex
- Better for detailed and hierarchical data

Key Difference

- Star → simplicity and speed
- Snowflake → complexity and flexibility

14. OLAP Operations (Page 9)

1. **Roll-up**
Aggregate data (city → country)
2. **Drill-down**
View detailed data

3. **Slice**
Focus on one dimension
4. **Dice**
Select multiple dimensions
5. **Pivot**
Rotate data view

15. Case Study Insight (Page 10)

Problem

- Stockouts (lost sales)
- Overstocking (waste)

Solution

1. Build **Data Warehouse**
2. Create **BI Dashboards**
3. Enable real-time tracking

Result

- Stockouts reduced by 75%
- Inventory costs reduced by 18%

16. Summary of Phases (Page 7 Table)

1. **Phase 1: Data Warehousing**
Build integrated data foundation
2. **Phase 2: Business Intelligence**
Reporting and dashboards
3. **Phase 3: Advanced Analytics**
Predictive modeling

MODULE 4 – DETAILED REVIEWER

Performance Management Frameworks and Systems

1. Performance Management Frameworks

Definition (Thorough Explanation)

Performance management is defined as a **systematic process that ensures employee activities and outputs are aligned with the organization's goals, helping the company gain a competitive advantage.**

It is described as:

- **continuous**, not a one-time evaluation
- **strategic**, aligned with long-term objectives
- focused on both **measurement and improvement**

Key Concept: The Logic Model (Page 2 Diagram)

A **Logic Model** is a **systematic tool used to illustrate the “if-then” causal chain between a program's resources, activities, and outcomes.**

According to the diagram on page 2, it shows:

- Inputs → Activities → Outputs → Outcomes → Impact

Purpose of the Logic Model

- links resources to results
 - ensures activities align with strategic goals
 - provides a shared understanding among stakeholders
-

2. Core Components of a Logic Model (Page 3)

1. Inputs (Resources)

Financial, human, and organizational resources required

2. Activities (Processes)

Actions performed using the inputs
Example: training, operations, programs

3. Outputs

Direct, measurable results of activities
Example: number of products produced

4. Outcomes

Changes or benefits resulting from outputs:

- short-term (skills, awareness)
- medium-term (behavior, adoption)
- long-term (sustainable change)

5. Assumptions and External Factors

Conditions that influence success but are outside control

Key Insight

The Logic Model provides a **cause-and-effect structure**, ensuring that:

- every activity contributes to desired outcomes
 - performance can be measured at each stage
-

3. Role of Logic Model in Business Performance Management

The Logic Model integrates:

- planning
 - implementation
 - evaluation
-

Key Functions

1. **Strategic Planning**
Aligns activities with mission and goals
 2. **Performance Management**
Provides measurable indicators (KPIs)
 3. **Communication**
Creates a common understanding among stakeholders
 4. **Evaluation and Accountability**
Enables tracking progress and measuring impact
 5. **Risk Identification**
Helps identify risks and plan mitigation strategies
-

4. Strategic Frameworks for Evaluation (Pages 4–5)

1. Balanced Scorecard

- Measures performance across multiple perspectives
 - Ensures holistic evaluation
-

2. Objectives and Key Results (OKRs)

- Links objectives with measurable key results
- Emphasizes transparency and alignment

3. SWOT Analysis

- Identifies:
 - Strengths
 - Weaknesses
 - Opportunities
 - Threats
-

4. PESTLE Analysis

- Examines external environment:
 - Political
 - Economic
 - Social
 - Technological
 - Legal
 - Environmental
-

5. Porter's Five Forces

- Analyzes industry competition:
 - rivalry
 - suppliers
 - buyers
 - substitutes
 - new entrants
-

6. Value Chain Analysis

- Examines internal processes to identify value creation and cost advantage

Key Insight

Each framework provides a **different perspective**, and organizations often combine them for comprehensive evaluation.

5. Designing Business Performance Management Systems

Definition (Page 5)

Designing a performance management system requires a **strategic, integrated, and continuous approach** that aligns:

- individual contributions
 - organizational goals
-

Key Steps in Design

1. Assess Organizational Needs and Goals

- Align system with mission and strategy
 - Identify purpose (accountability, development, etc.)
 - Identify existing gaps
-

2. Establish Clear Goals and Metrics

- Cascade goals from organizational to individual level
 - Use:
 - SMART goals
 - OKRs
 - Define measurable KPIs
-

3. Implement Continuous Feedback Cycle

- Replace annual reviews with ongoing feedback
 - Encourage open communication
 - Use multiple feedback sources
-

4. Integrate Development and Rewards

- Provide training and coaching
 - Align rewards with performance
 - Use Performance Improvement Plans (PIPs)
-

5. Leverage Technology

- Use systems to automate tracking
 - Apply analytics for decision-making
-

6. Monitor, Evaluate, and Adapt

- Continuously assess effectiveness
 - Adjust based on:
 - data
 - feedback
 - changing business needs
-

6. Continuous Improvement in Business Evaluation

Key Idea (Page 7)

A performance system succeeds when it **triggers improvement**, not just measurement.

Hoshin Kanri Framework

Also called:

- **policy deployment**
 - **compass management**
-

Purpose

- Align organization around key strategic goals
 - Bridge gap between strategy and execution
-

X-Matrix (Page 7 Diagram)

Visual tool showing alignment between:

- long-term goals
 - annual objectives
 - key initiatives
 - performance metrics
-

7. Obeya (“Big Room”) Framework (Page 8)

Definition

A **visual management framework** that creates a shared workspace for decision-making and collaboration.

Four Walls of Obeya

1. **Strategy Wall**
Displays mission, vision, and goals
2. **Delivery Wall**
Tracks projects and progress
3. **Performance Wall (Metrics Wall)**
Displays KPIs and performance indicators

4. Improvement Wall

Focuses on continuous improvement

Key Insight

Obeya supports:

- collaboration
 - transparency
 - alignment
-

8. Monthly Business Reviews (MBR) (Page 9)

Definition

Structured meetings where leaders:

- assess performance
 - review metrics
 - plan improvements
-

Two Main Sections

1. **Look Back (Review and Reflect)**
Analyze past performance
Review KPIs and actions
 2. **Look Forward (Plan and Commit)**
Set future goals
Identify risks
allocate resources
-

Key Insight

MBR creates a balance between:

- accountability
 - learning
 - strategic execution
-

9. KPI Dictionary

Definition

A centralized repository of standardized KPI definitions.

Purpose

- ensures consistency
 - improves transparency
 - aligns understanding across teams
-

10. Continuous Improvement Framework

Key Insight (Page 9)

Without continuous improvement:

- evaluation becomes “autopsy” (too late to fix problems)

With CI:

- evaluation becomes **diagnostic tool for real-time correction**
-

11. Case Study Insight (Pages 10–11)

Problem

- low visibility
 - operational inefficiency
 - data silos
 - fragmented metrics
-

Solution Approach

1. Digital Strategy & Solutions

- tracking systems
- cloud platforms

2. Performance Management Redesign

- Balanced Scorecard or OKRs
- KPI Dictionary

3. Implementation & Governance

- phased rollout
 - governance mechanisms (MBR, Obeya)
-

Expected Outcomes

- improved efficiency
- better customer satisfaction
- stronger decision-making

Reviewer: Foundations of Outsourcing and Offshoring

1. Definition & Distinction

Outsourcing

Outsourcing refers to the practice of **contracting specific business or IT functions to an external service provider**. The provider may be **in the same country or another country**.

Organizations outsource tasks so they can **focus on their core competencies** while gaining **specialized expertise and cost efficiency**.

Commonly outsourced IT services:

- Software development
- Technical support
- System maintenance
- Data processing
- Cloud services

Offshoring

Offshoring involves **relocating business or IT operations to another country**.

It can be done in two ways:

1. **Through a third-party vendor** (offshore outsourcing)

2. **Through a company-owned subsidiary** (captive offshoring)

Companies offshore mainly because of:

- Lower labor costs
- Access to global talent
- 24/7 operations

2. Examples of Outsourcing and Offshoring

Outsourcing

Outsourcing means **another company performs the work instead of the company's own employees**.

The provider can be **local or international**.

Example (Local Outsourcing)

A Philippine bank hires a **local IT company in Manila** to maintain its information system and provide technical support.

Example (International Outsourcing)

A company in Cebu hires a **software firm in India** to develop a mobile application.

Offshoring

Offshoring focuses on **moving work to another country**, whether the company owns the operation or hires another company.

Example (Company-Owned Offshoring)

A Philippine IT company opens its **own software development office in Vietnam** to benefit from skilled developers and lower labor costs.

Example (Offshore Outsourcing)

A **U.S. company hires a BPO company in the Philippines** to manage customer support systems.

3. Historical Context

Outsourcing became popular in the **late 20th century** when companies wanted to **reduce operational costs and improve efficiency**.

Early outsourcing focused on **non-core functions**, such as:

- Payroll
- Data entry

With the development of **information and communication technologies (ICT)**, outsourcing expanded to **IT services**.

Offshoring developed together with **globalization and international trade**.

Technologies that helped offshoring grow include:

- Internet connectivity
- Cloud computing

- Project management tools

Countries like **India, the Philippines, and Eastern Europe** became major hubs because of:

- Skilled workforce
- Lower labor costs
- Strong English proficiency

4. Key Drivers of IT Outsourcing and Offshoring

1. Cost Reduction and Operational Efficiency

Companies outsource or offshore to **lower operating costs**.

Maintaining an in-house IT team requires:

- Salaries
- Benefits
- Training
- Infrastructure

Benefits

- Reduced labor costs
- Pay only for needed services
- More efficient operations

Example:

A company outsources software testing to a country with lower labor costs.

2. Access to Specialized Skills and Global Talent

Some organizations **do not have advanced IT skills locally**.

Outsourcing and offshoring allow access to **global experts**.

Benefits

- Experts in cybersecurity, AI, cloud computing, and data analytics
- Faster project completion
- Less training for internal staff

Example:

A company hires an offshore team specializing in **cloud migration**.

3. Focus on Core Business Activities

Outsourcing **non-core IT functions** allows companies to focus on their **main business goals**.

Advantages

- Management focuses on strategy and innovation
- Experts handle IT support
- Higher productivity

Example:

A retail company outsources IT helpdesk services to focus on **customer experience and sales**.

4. Scalability and Flexibility of IT Resources

Outsourcing and offshoring allow companies to **adjust IT resources based on business needs**.

Benefits

- Increase resources during peak demand
- Reduce resources during low demand
- No need to hire permanent staff

Example:

A startup hires extra developers during a

product launch and reduces the team afterward.

5. Technological Advancements and Digital Platforms

Modern technologies make outsourcing and offshoring **easier and more efficient**.

Important technologies

- Cloud computing
- Collaboration platforms
- Secure data-sharing systems

Example:

Global teams work together using **GitHub, Jira, and cloud servers**.

6. Competitive Pressure in Global Markets

Companies must **innovate quickly and reduce costs** to stay competitive.

Outsourcing and offshoring help by improving:

- Speed
- Quality
- Efficiency

Benefits

- Faster time-to-market
- Ability to compete globally
- Better service delivery

Example:

A software company outsources development to **release products faster than competitors**.

5. Strategic Fit

Strategic fit means outsourcing and offshoring decisions must **align with the organization's overall strategy and long-term goals**.

Managers must identify:

- **Core functions** – strategic and value-creating activities that should remain in-house
- **Non-core functions** – supportive activities that can be outsourced

If outsourcing does not fit the strategy, it may cause:

- Loss of control

- Quality issues
- Data security risks
- Vendor dependency

6. Strategic Considerations

1. Alignment with business goals and digital strategy

Outsourcing should support the company's **long-term objectives and innovation plans**.

2. Risk management (data security, continuity, compliance)

Organizations must manage risks such as:

- Data privacy
- Cybersecurity
- Service disruptions
- Legal compliance

3. Cost-benefit analysis beyond short-term savings

Companies should evaluate **long-term costs, service quality, and sustainability**, not only immediate savings.

4. Impact on employees and organizational knowledge

Outsourcing may affect:

- Employee morale
- Skill retention
- Loss of institutional knowledge

5. Ethical, legal, and social responsibility implications

Organizations must ensure:

- Fair labor practices
- Legal compliance
- Environmental responsibility
- Protection of company reputation

Reviewer: The Outsourcing Lifecycle and Sourcing Strategy

1. The Sourcing Decision Process: To Build or To Buy?

Before outsourcing, organizations must first ask:

“Should we even do this?”

This decision is based on **three critical pillars**:

1. Make-or-Buy Analysis
2. Feasibility Studies
3. Risk Assessment

These help companies decide whether outsourcing is the **best option**.

2. Make-or-Buy Analysis

Make-or-Buy Analysis is a **management accounting and strategic decision-making tool** used to determine whether a task should be:

- **Make** – done internally by company employees
- **Buy** – outsourced to an external supplier

The goal is to choose the option that provides the **best balance of cost, quality, control, and strategic value**.

This analysis considers:

- Financial costs

- Operational capabilities
 - Risks
 - Long-term business impact
-

Make (In-house)

The organization performs the task internally.

Advantages

- More control over quality
 - Better protection of intellectual property (IP)
 - Strong alignment with company culture
-

Buy (Outsource)

The organization hires an external vendor to perform the task.

Advantages

- Lower operational costs
- Access to specialized skills and expertise

- Faster scaling of operations

3. Feasibility Studies

Before signing a contract, organizations must perform a **feasibility study**.

This determines whether outsourcing is **technically and operationally possible**.

Feasibility studies mainly focus on:

1. Technical Feasibility
2. Operational Feasibility

Technical Feasibility

Technical feasibility checks whether the vendor has the **technology, systems, and technical expertise** needed.

Example considerations:

- Does the vendor's system integrate with existing systems?
- Does the vendor have the required technology stack?

If the vendor lacks technical capability, the outsourcing project may fail.

Operational Feasibility

Operational feasibility examines whether the organization can **manage and support the outsourcing arrangement**.

Example considerations:

- Can the company coordinate with teams in different time zones?
- Can managers supervise offshore teams effectively?

Operational feasibility ensures that **daily operations will run smoothly**.

4. Risk Assessment

Outsourcing does not eliminate risks. Instead, organizations must **identify and manage risks**.

The three major risks are:

1. Financial Risk
2. Operational Risk
3. Compliance Risk

Financial Risk

Financial risk evaluates whether the vendor is **financially stable** throughout the contract period.

Key considerations:

- Vendor solvency – Is the vendor financially stable?
- Cash flow – Can the vendor continue investing in staff and infrastructure?
- Dependency risk – What happens if the vendor shuts down or is acquired?

Ignoring financial risk may cause:

- Service interruptions
- Emergency vendor replacement
- Additional costs

Operational Risk

Operational risk focuses on possible **disruptions in daily service delivery**.

Key considerations:

- Infrastructure reliability (internet, power, data centers)
- Business continuity and disaster recovery
- Service-level resilience after system failures

Operational risk can affect:

- Productivity
- Customer experience
- Revenue

Therefore, **backup plans and recovery systems are essential**.

Compliance Risk

Compliance risk evaluates whether the vendor follows **legal and regulatory requirements**.

Key considerations:

- Compliance with data privacy laws (e.g., GDPR)
- Security controls and regular audits
- Lawful handling of cross-border data transfers

Failure to manage compliance risk may lead to:

- Legal penalties

- Reputational damage
- Loss of customer trust

5. Defining Scope and Requirements: The Blueprint

Translating business needs into specification

After deciding to outsource, organizations must **clearly define the scope and requirements**.

Business stakeholders usually describe **outcomes**, while IT managers must convert them into **technical specifications**.

Example:

- Stakeholder: “I want the app to be faster.”
- IT Specification: “The system must support 10,000 concurrent users with latency below 200ms.”

Requirements are divided into:

1. Functional Requirements
2. Non-Functional Requirements

Functional Requirements

Functional requirements describe **what the system must do**.

They define system features, processes, and user interactions.

Examples:

- The portal must allow users to reset passwords
- The system must generate monthly reports automatically
- Users must be able to submit and track requests online

These requirements ensure the system provides the **necessary capabilities**.

Non-Functional Requirements

Non-functional requirements describe **how the system should behave and perform**.

They focus on the **quality and performance of the system**.

Common categories include:

Performance

Response time, processing speed, and system capacity.

Security

Authentication, authorization, encryption, and access control.

Scalability

Ability to handle increased users or transactions.

Availability and Reliability

System uptime, backup systems, and disaster recovery.

Usability

Ease of use and user experience.

- Broad and high-level questions

- Non-binding (no pricing commitments)

6. The Vendor Selection Process: Finding “The One”

Once requirements are defined, the organization must **select the right vendor**.

The vendor selection process involves:

1. RFI vs RFP
2. Due Diligence
3. Evaluation Criteria

- Sent to many vendors (usually 10–20)

It helps organizations **filter vendors before deeper evaluation**.

RFP (Request for Proposal)

An **RFP** is used after shortlisting vendors.

Purpose:

- Request detailed proposals and pricing

7. RFI vs RFP

RFI (Request for Information)

An **RFI** is used in the **early stage of vendor selection**.

Purpose:

- Understand the market
- Identify vendors with the required capability

Characteristics:

- Includes detailed requirements
- Vendors submit pricing and commercial terms
- Vendors propose technical solutions

RFPs are usually sent to **3–5 shortlisted vendors**.

Characteristics:

8. Due Diligence

Due diligence ensures that the vendor is **reliable and capable**.

Organizations verify vendor claims through:

Site Visits

Visiting the vendor's office or facilities.

Reference Checks

Talking to the vendor's previous or current clients.

Financial Audits

Checking the vendor's financial stability for long-term partnership.

9. Evaluation Criteria

Organizations often use a **Weighted Scoring Model** to choose the best vendor.

Vendors are evaluated using criteria such as:

- Cost (30%)
- Technical Expertise (40%)
- Cultural Fit (20%)
- Geographic Location (10%)

Each vendor receives scores for every category.

10. Weighted Scoring Model

In a **Weighted Scoring Model**, each evaluation criterion is assigned a **weight based on importance**.

Process:

1. Assign a weight (percentage) to each criterion.
2. Score vendors for each criterion.
3. Multiply scores by the assigned weights.
4. Add all scores to get the final ranking.

This method ensures that **important factors have greater influence in the final decision**.

Reviewer: Contract Management and Legal Frameworks

1. Key Contractual Components: The "Teeth" of the Agreement

Contracts in outsourcing must clearly define **rules, responsibilities, and expectations** between the client and the vendor.

These components are called the **"teeth" of the agreement** because they **enforce accountability**.

The main components include:

1. Service Level Agreements (SLAs)
2. Key Performance Indicators (KPIs)

3. Exit Clauses

2. Service Level Agreements (SLAs)

A **Service Level Agreement (SLA)** defines the **minimum acceptable level of service** that the vendor must provide.

It sets clear performance standards that the vendor must meet.

Example:

- *“The system must be available 99.9% of the time.”*

If the vendor fails to meet the SLA, they usually must give **Service Credits**, which are **penalties or discounts on the next payment**.

Purpose of SLAs:

- Ensure service quality
- Protect the client
- Hold vendors accountable

3. Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) are **measurable values used to evaluate how well the vendor performs their work**.

While **SLAs measure service availability**, **KPIs measure performance quality**.

Examples:

- Average time to resolve a support ticket
- Customer satisfaction score
- System response time

Purpose of KPIs:

- Monitor vendor performance
- Improve service quality
- Support continuous improvement

4. Exit Clauses

An **Exit Clause** (also called a **Termination Clause**) explains **how the contract can be ended**.

Since outsourcing relationships may eventually end, the contract must clearly define the exit process.

Exit clauses usually include:

- How the vendor returns the client's data
- Transition support during vendor replacement

- Costs for ending the contract early

Without an exit clause, organizations may experience **Vendor Lock-in**, where they become **dependent on a vendor and cannot easily switch providers**.

Quick Definitions

SLA

A written agreement that defines the **minimum acceptable service level and penalties if not met**.

KPI

A measurable value used to **evaluate vendor performance**.

Exit Clause

A contract section explaining **how the agreement can be terminated**, including transition support and data return.

5. Pricing Models: How Do We Pay?

Pricing models determine **how the client pays the vendor**.

Choosing a pricing model is mainly about **Risk Allocation**, meaning **who takes the risk if the project takes longer or costs more than expected**.

Fixed-Price

How it Works

- A **set price is agreed for a specific scope of work**.

Who carries the Risk

- The **Vendor**

Best Used For

- Small projects
 - Projects with **clear and stable requirements**
-

Time & Materials (T&M)

How it Works

- The client pays for **actual hours worked and expenses**.

Who carries the Risk

- The **Client**

Best Used For

- **Agile projects**
 - Projects where requirements change often
-

Cost-Plus

How it Works

- The client pays the vendor's **actual costs plus a fixed profit percentage**.

Who carries the Risk

- The **Client**

Best Used For

- Long-term partnerships
- Captive centers or internal offshore teams

Gain-Sharing

How it Works

- The vendor receives a **bonus if they help the client save money or achieve major goals**.

Who carries the Risk

- **Both the Client and Vendor**

Best Used For

- Strategic innovation projects
- Long-term performance partnerships

6. Legal and Regulatory Considerations: Staying Out of Trouble

Outsourcing contracts must follow **legal and regulatory requirements** to avoid penalties, disputes, or security problems.

Important legal areas include:

1. Data Privacy
2. Intellectual Property (IP) Rights
3. Jurisdiction

7. Data Privacy (The GDPR Effect)

Companies remain responsible for **protecting customer data**, even if a vendor handles the system.

Example:

If a **European company outsources to the Philippines**, the company must still follow **GDPR (General Data Protection Regulation)**.

This means:

- The vendor must have strong security systems
- Data privacy laws must be followed
- Data breaches can lead to **large fines**

A security failure at the vendor can still **damage the client company.**

8. Intellectual Property (IP) Rights

Intellectual Property (IP) Rights determine **who owns the products or systems created by the vendor.**

Example:

If a vendor builds a custom application for a client, the contract must clearly state that **the IP belongs to the client.**

Without this clause:

- The vendor might reuse or sell the software
- Competitors could gain access to the same solution

Therefore, contracts must clearly specify **ownership of code, software, and innovations.**

9. Jurisdiction

Jurisdiction defines **which country's laws apply to the contract and where disputes will be resolved.**

Two important parts:

Governing Law

Determines **which country's legal system will interpret the contract.**

Venue

Determines **where legal disputes or arbitration will take place.**

Many international outsourcing contracts use **neutral jurisdictions**, such as:

- Singapore
- London
- New York

This ensures fairness for both parties.

Reviewer: Vendor & Relationship Management

1. Building a Strategic Partnership: Moving Beyond "Us vs. Them"

In outsourcing, the goal is not just to hire a vendor to complete tasks.

Organizations should aim to build a **Strategic Partnership.**

A **Strategic Partnership** is a **long-term relationship between two organizations where both work together to achieve shared goals and mutual success.**

Instead of treating the vendor as just a service provider, the client and vendor **collaborate to improve business results.**

Characteristics of a Strategic Partnership

Shared Goals

Both the client and vendor benefit when the project succeeds.

Example:

A vendor may receive a **bonus for helping**

the client increase user engagement by 20%.

This motivates the vendor to contribute to the client’s business success.

Transparency

Both parties should **share plans and important information openly**.

Example:

If the client plans a **major product launch in six months**, the vendor should know early so they can **prepare and scale their team**.

Transparency helps avoid delays and miscommunication.

Investment in Innovation

A strategic vendor does more than just complete assigned tasks.

Instead of only fixing problems, the vendor may:

- Suggest new technologies
- Recommend improvements
- Propose new features

This helps the client **improve business performance and innovation**.

Traditional Vendor vs Strategic Partnership

Traditional Vendor	Strategic Partnership
Just completes tasks	Helps achieve business success
Limited communication	Open communication
Fixes problems only	Suggests improvements
Short-term focus	Long-term relationship

2. Governance Structures: The “Rule of Engagement”

When two large organizations work together, there must be a **Governance Structure**.

A **Governance Structure** is a **formal framework of roles, committees, and communication rules that manage the outsourcing relationship**.

It ensures that:

- Decisions are organized
- Responsibilities are clear
- Problems are handled properly

Key Components of Governance Structures
Defined Roles and Responsibilities

Each person must clearly understand their **duties and responsibilities**.

Example roles:

- Project Manager
- Team Leader
- Executive Sponsor

This prevents confusion and overlapping responsibilities.

Committees

Committees are groups responsible for **decision-making and relationship management**.

Examples:

Steering Committee

- Handles major decisions
- Resolves serious issues

Operational Committee

- Manages daily operations
 - Monitors ongoing activities
-

Communication Protocols

Communication Protocols define **how communication happens between the client and vendor**.

They specify:

- Who communicates
- How often communication occurs
- What tools are used

Examples:

- Weekly meetings
 - Monthly reports
-

Escalation Process

The **Escalation Process** is a step-by-step method for **raising unresolved problems to higher authority**.

Example escalation path:

Employee → Manager → Director → Executive

This ensures problems are solved quickly and efficiently.

3. The Three-Tier Governance Model

Effective outsourcing relationships often follow a **Three-Tier Governance Model**.

Operational Level

Focus: Daily work and tasks.

People involved:

- Developers
- Project Managers

Frequency:

Daily or weekly meetings.

Purpose:

- Manage daily tasks
- Fix bugs
- Track progress

Tactical Level

Focus: Performance management.

People involved:

- Managers
- Department Heads

Frequency:

Monthly meetings.

Purpose:

- Review KPIs
- Review SLAs
- Evaluate performance trends

Strategic Level

Focus: Long-term planning.

People involved:

- Executives
- CEOs
- Vice Presidents

Frequency:

Quarterly or bi-annual meetings.

Purpose:

- Discuss long-term strategy
 - Make major decisions
 - Review contract renewals
-

4. Communication Protocols and Escalation Paths

In outsourcing, communication must be **clear and structured**, especially during urgent situations.

Communication Protocol

A **Communication Protocol** is a document that defines **how teams communicate**.

It includes:

- Communication tools
- Frequency of meetings
- Communication responsibilities

Examples:

- Jira for task tracking
 - Slack for quick discussions
 - Email for formal approvals
-

Escalation Path

An **Escalation Path** defines **who should handle a problem if it cannot be resolved at a lower level**.

Example:

If a problem is not solved within **2 hours**, it is escalated to the **next level of authority**.

Example escalation:

Manager → Director → Executive

This prevents **delayed or unresolved issues**.

5. Managing Cultural Differences: The Invisible Challenge

In global outsourcing, cultural differences can create **communication misunderstandings**.

Even when everyone speaks English, differences in communication style and workplace culture may cause problems.

6. High-Context vs Low-Context Communication

Different cultures communicate in different ways.

Low-Context Communication

Common in countries like:

- USA
- Germany

Characteristics:

- Communication is **direct and literal**

- Messages are clearly stated

Example:

If someone says **“No”**, it directly means **“No.”**

High-Context Communication

Common in countries like:

- Philippines
- Japan
- India

Characteristics:

- Communication is **indirect and polite**
- Meaning may depend on tone or context

Example:

A vendor might say **“That might be difficult”** as a polite way of saying **“No.”**

If the client misunderstands this, they might think the task is still possible.

7. Power Distance

Power Distance refers to how comfortable employees are with **challenging authority or disagreeing with leaders**.

In some cultures, employees may avoid disagreeing with:

- Managers
- Clients
- Executives

This can be dangerous in IT projects because developers may **not report problems early**.

Solution: Psychological Safety

Managers should create a **psychologically safe environment** where vendors feel comfortable:

- Raising risks
- Reporting problems
- Suggesting improvements

This helps prevent project failures.

8. Time-Zone and Holiday Awareness

In global outsourcing, teams often work in **different time zones and countries**.

Managers must consider:

- National holidays
- Working hours
- Cultural schedules

Example:

If a client schedules a sprint during a **national holiday in the vendor's country**, work may be delayed.

A good solution is to maintain a **Joint Calendar** that includes:

- Holidays from both countries
- Important project deadlines
- Sprint schedules

This helps maintain **coordination and work-life balance**.

Reviewer: Financial Management and Cost Optimization

1. Total Cost of Ownership (TCO): Peeling the Onion

Total Cost of Ownership (TCO) refers to the **total cost of outsourcing**, including both **visible costs and hidden costs**.

The concept is called **“Peeling the Onion”** because costs appear in layers:

- The **outer layer** contains visible costs.
- The **inner layers** contain hidden or indirect costs.

Organizations must consider **all layers of cost** to understand the real financial impact of outsourcing.

2. Direct Costs (The Visible Part)

Direct Costs are expenses **clearly written in the outsourcing contract**.

These costs are usually easy to identify because they appear as **line items in the agreement**.

Examples of Direct Costs:

Vendor Fees

Payments made to the vendor for services.

Examples:

- Monthly retainers
- Hourly service rates

Setup Costs

Costs needed to start the outsourcing arrangement.

Examples:

- Software licenses
- Hardware purchases
- Initial training

Transition Costs

Costs during the **handover period** when work is transferred from the internal team to the vendor.

This is often called the “**double-bubble**” **period**, because the company is paying for:

- The **internal team**
- The **vendor team**

at the same time.

3. Indirect and Hidden Costs (The “Underwater” Part)

Indirect and Hidden Costs are expenses that **are not always included in the contract** but still affect the organization.

These costs often surprise management because they are **not immediately visible**.

Governance & Management

Even with outsourcing, the organization still needs **internal managers to oversee the vendor relationship**.

Typical estimate:

For every **\$1 million spent on outsourcing**, organizations may spend **5%–10% on internal management**.

These managers handle:

- Vendor communication
- Performance monitoring
- Contract management

Knowledge Transfer

Knowledge Transfer happens when internal experts **train the vendor’s team** about systems and processes.

During this time:

- Internal staff stop their regular work
- Productivity may temporarily decrease

This creates a **hidden cost due to lost productivity**.

Quality Gap Costs

These costs occur when the vendor **makes mistakes or does not fully understand the business.**

Examples:

- Fixing incorrect work
- Rewriting software
- Resolving operational errors

These extra corrections increase the overall cost.

4. Cost Monitoring and Benchmarking: Keeping the Vendor Honest

Organizations must continuously monitor vendor costs to ensure they remain **competitive and fair.**

Benchmarking

Benchmarking is the process of **comparing the vendor's price and performance with the current market standards.**

This helps determine if the vendor's services are still **reasonable and competitive.**

Some contracts include a **Benchmarking Clause.**

A **Benchmarking Clause** allows the client to hire a **third-party evaluator** (such as Gartner or IDC) to compare vendor rates with market prices.

If market prices become significantly lower, the vendor may be required to **reduce their rates during the contract period.**

5. Realized Savings vs Expected Savings

Organizations often compare **expected benefits** with **actual results.**

Hard Savings

Hard Savings are **actual financial savings.**

Example:

If outsourcing reduces expenses from **\$1M to \$800K**, the company saves **\$200K.**

This is measurable in cash.

Soft Savings

Soft Savings are **non-cash benefits.**

Examples:

- Time saved
- Increased efficiency
- Avoided future costs

Soft savings improve productivity but may be harder to measure.

Baseline Comparison

To measure savings accurately, organizations use **Baseline Comparison.**

This technique records **how much the company was spending before outsourcing**.

Then they compare it to the **cost after outsourcing** to verify if the outsourcing decision truly saved money.

6. Billing and Invoice Verification: The Audit Trail

Before paying a vendor invoice, companies must verify the charges using an **Audit Trail**.

This ensures the organization **only pays for legitimate work and services**.

7. The Three-Way Match

The **Three-Way Match** is a verification process used before paying vendor invoices.

It compares **three important documents**.

1. The Contract

Checks if the vendor is charging the **correct rates agreed in the contract**.

Example:

Are the hourly or monthly fees correct?

2. The Timesheet / Work Log

Confirms whether the vendor **actually completed the work they are billing for**.

Examples:

- Number of hours worked

- Tasks completed

- Story points delivered
-

3. The Performance (SLA) Report

Verifies if the vendor met the **Service Level Agreements (SLAs)**.

If the vendor failed to meet SLA targets, the invoice should include **Service Credits (penalties or discounts)**.

8. Performance-Based Billing

Modern outsourcing contracts often use **Performance-Based Billing**.

In this model, payments depend on **vendor performance**.

Example:

- If the vendor achieves **100% of KPIs**, they receive full payment.
- If the vendor achieves **80% of KPIs**, the payment is reduced.

This system encourages vendors to **maintain high service quality and performance**.

MODULE 1: STATISTICS AND DATA

Nature of Statistics

Statistics

- **Definition:** The science of conducting studies to **collect, organize, summarize, analyze, and draw conclusions from data**.
- Used in: health, education, business, sports, research, etc.

Variables and Data

Variable

- A **well-defined characteristic or attribute** that can assume different values.

Data

- The **values (measurements or observations)** that variables can assume.

Example:

- Variable: Gender → Data: Male, Female
- Variable: Age → Data: 14, 15, 16

Types of Statistics

1. Descriptive Statistics

- **Definition:** Collection, organization, summarization, and presentation of data.
- Focus: **describes the whole group (population)**

2. Inferential Statistics

- **Definition:** Generalizing from samples to populations.
- Includes:
 - Estimation
 - Hypothesis testing
 - Prediction
- Uses **probability (chance of an event occurring)**

Population and Sample

Population

- All subjects being studied.

Sample

- A group selected from the population.

Types of Variables

1. Qualitative Variables

- Non-numerical categories
- Example: Gender, Religion

2. Quantitative Variables

- Numerical values

Types:

- **Discrete** → Countable values
Example: number of students
- **Continuous** → Infinite values (measured)
Example: height, temperature

Measurement Scales

Nominal Level

- No ranking or order
- Example: religion, zip codes

Sampling Methods

Systematic Random Sampling

- Select every kth subject

Stratified Random Sampling

- Divide population into groups (strata), then sample

Cluster Random Sampling

- Divide into clusters, randomly select clusters

Research Design

Observational Study

- No intervention
- Only observes data

Experimental Study

- Researcher **manipulates variables**

Variables in Experimental Study

- **Independent Variable** → manipulated variable
- **Dependent Variable** → outcome variable

Groups

- **Treatment Group** → receives treatment
 - **Control Group** → does not receive treatment
-

Important Terms

Confounding Variable

- A variable that affects results but cannot be separated

Placebo

- A **fake treatment** with no effect
-

Five Principles of Experimental Design

1. **Control** → keep conditions similar
 2. **Randomize** → reduce bias
 3. **Replicate** → repeat experiment
 4. **Block** → group similar subjects
 5. **Blinding** → hide treatment information
-

MODULE 2: DESCRIBING THE DATA

Raw Data

- Data in its **original form**
-

Three Rules of Data Analysis

1. Make a picture
2. Make a picture
3. Make a picture

Graphs help reveal patterns and trends.

Frequency Distribution

Definition

- Organizing data into **classes and frequencies**

Types:

- **Categorical**
- **Grouped**

$$\bar{X} = \frac{\sum X}{n}$$

Graphs

- **Bar Chart** → compares categories
- **Pie Chart** → shows proportions

Grouped Frequency Distribution Steps

1. Find **highest (H) and lowest (L)**
2. Compute **Range**

Formula: Range

$$R = H - L$$

3. Choose number of classes
4. Compute class width

Formula: Class Width

$$\text{Width} = \frac{R}{\text{number of classes}}$$

5. Create classes
6. Tally data
7. Count frequencies

Stem and Leaf Plot

- Displays data while **keeping original values**

MEASURES OF CENTRAL TENDENCY

1. Mean (Average)

Formula:

- Sum of all values divided by total number

2. Median

- Middle value of ordered data

Notes:

- Odd → exact middle
- Even → average of two middle values

3. Mode

- Most frequent value

Statistic vs Parameter

- **Statistic** → from sample
- **Parameter** → from population

MEASURES OF VARIABILITY

Variance and Standard Deviation

Population Variance

Formula:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

Population Standard Deviation

Formula:

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

Sample Variance

Formula:

$$s^2 = \frac{\sum(X - \bar{X})^2}{n - 1}$$

Sample Standard Deviation

Formula:

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

MEASURES OF POSITION

Percentiles

- Divide data into **100 equal groups**

Formula: Percentile Rank

$$\text{Percentile} = \frac{(\text{number below } X) + 0.5}{\text{total values}} \times 100$$

Formula: Position of Percentile

$$c = \frac{n \cdot p}{100}$$

Quartiles

- Divide data into **4 groups**
- **Q1 = 25th percentile**
- **Q2 = Median (50th percentile)**

- **Q3 = 75th percentile**

Interquartile Range (IQR)

Formula:

$$IQR = Q3 - Q1$$

Deciles

- Divide data into **10 groups**

OUTLIERS

Definition

- Extremely high or low values

Steps:

1. Find Q1 and Q3
2. Compute IQR
3. Multiply by 1.5
4. Check limits:

Outlier Condition

$$< Q1 - 1.5(IQR) \text{ or } > Q3 + 1.5(IQR)$$

EXPLORATORY DATA ANALYSIS (EDA)

Five-Number Summary

1. Minimum
2. Q1
3. Median
4. Q3
5. Maximum

Boxplot

- Graph showing distribution using five-number summary

Interpretation of Boxplot

- Median centered → symmetric
- Median left → positively skewed
- Median right → negatively skewed

TRADITIONAL VS EDA

Traditional	EDA
Mean	Median
Standard Deviation	IQR
Histogram	Boxplot

Frequency Distribution Stem-and-leaf

MODULE 3: NORMALITY AND ESTIMATION

NORMAL DISTRIBUTION

What is Normal?

- A **normal distribution** is used to determine **normal intervals** in real-life data such as blood pressure, cholesterol, and other measurements.
- It helps researchers determine whether values are **within the normal range or not**.

SHAPE OF DISTRIBUTION

Symmetric Distribution

- Data are evenly distributed about the mean.
- A **normal distribution is symmetric**.

Skewed Distribution

Negatively Skewed (Left-Skewed)

- Tail is on the left.
- $\text{Mean} < \text{Median} < \text{Mode}$

Positively Skewed (Right-Skewed)

- Tail is on the right.
- $\text{Mean} > \text{Median} > \text{Mode}$

NORMAL DISTRIBUTION

Definition

- A **continuous, symmetric, bell-shaped distribution of a variable**.

PROPERTIES OF A NORMAL DISTRIBUTION

- Bell-shaped curve
- $\text{Mean} = \text{Median} = \text{Mode}$
- Unimodal (one peak)
- Symmetric about the mean
- Continuous (no gaps)
- Curve never touches the x-axis
- Total area = 1 or 100%

Empirical Rule (Very Important)

- 68% within 1 standard deviation
 - 95% within 2 standard deviations
 - 99.7% within 3 standard deviations
-

NORMAL DISTRIBUTION FORMULA

Important Formula:

$$y = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$$

Where:

- $e \approx 2.718$
- $\pi \approx 3.14$
- μ = population mean
- σ = population standard deviation

STANDARD NORMAL DISTRIBUTION

Definition

- A normal distribution with:
 - Mean = 0
 - Standard deviation = 1

Z-SCORE (STANDARD SCORE)

Important Formula:

$$z = \frac{\text{value} - \text{mean}}{\sigma}$$

- Converts any normal variable into standard normal form

FINDING PROBABILITIES USING Z

Cases:

1. **Left of z**
 - Use table directly
2. **Right of z**
 - $1 - (\text{table value})$

3. Between two z values

larger area – smaller area

Important Note:

- $P(a \leq z \leq b) = P(a < z < b)$
- Probability of exact value = 0 (continuous distribution)

APPLICATIONS OF NORMAL DISTRIBUTION

Steps:

1. Convert value to z-score
2. Use z-table
3. Interpret probability

CENTRAL LIMIT THEOREM (CLT)

Definition

- As sample size increases, the distribution of sample means becomes **approximately normal**.

SAMPLING DISTRIBUTION OF THE MEAN

Definition

- Distribution of all possible sample means

PROPERTIES

1. Mean of sample means = population mean

$$\mu_{\bar{x}} = \mu$$

2. Standard deviation of sample means (Standard Error)

Important Formula:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Z-FORMULA FOR SAMPLE MEAN

Important Formula:

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

WHEN TO USE CLT

- If population is normal → any sample size
 - If not normal → use $n \geq 30$
-

WHICH FORMULA TO USE

- **Individual value:**

$$z = \frac{X - \mu}{\sigma}$$

- **Sample mean:**

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

ESTIMATION

Definition

- The process of estimating a **population parameter** using a sample
-

POINT ESTIMATE

Definition

- A **single value estimate** of a parameter
- Best estimate of population mean:

$$\bar{X}$$

PROPERTIES OF A GOOD ESTIMATOR

1. **Unbiased** → equals true value on average
 2. **Consistent** → improves as sample size increases
 3. **Efficient** → smallest variance
-

CONFIDENCE INTERVALS

Definition

- A **range of values** used to estimate a parameter
-

CONFIDENCE LEVEL

- Probability that interval contains the true parameter

Examples:

- 95% → $\alpha = 0.05$
 - 99% → $\alpha = 0.01$
-

CRITICAL VALUE

- $z_{\alpha/2}$ corresponds to confidence level

$$\bar{X} \pm t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

CONFIDENCE INTERVAL (σ known or $n \geq 30$)

Important Formula:

$$\bar{X} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

MAXIMUM ERROR OF ESTIMATE

Formula:

$$E = z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

STUDENT t-DISTRIBUTION

Used when:

- σ is unknown
- $n < 30$

PROPERTIES OF t-DISTRIBUTION

- Bell-shaped
- Symmetric
- Mean = 0
- Depends on **degrees of freedom (df = n - 1)**
- Approaches normal distribution as n increases

CONFIDENCE INTERVAL (σ unknown)

Important Formula:

DEGREES OF FREEDOM

Formula:

$$df = n - 1$$

Excel Formulas

Measure	Excel Formula
Mean	=AVERAGE(A2:A101)
Median	=MEDIAN(A2:A101)
Mode	=IFERROR(MODE.SNGL(A2:A101),"No Mode")
Std Dev (Sample)	=STDEV.S(A2:A101)
Std Dev (Population)	=STDEV.P(A2:A101)
Variance (Sample)	=VAR.S(A2:A101)
Variance (Population)	=VAR.P(A2:A101)
Percentile	=PERCENTILE.INC(A2:A101,k)
Quartile	=QUARTILE.INC(A2:A101,n)
IQR	=Q3-Q1

Z-score **(X-Mean)/StdDev** **99%** **2.576**

RANDOM
=RAND()

Sample size
=COUNT(A2:A51)

Q1 (First Quartile)
=QUARTILE.INC(A2:A101,1)

Q2 (Median)
=QUARTILE.INC(A2:A101,2)

Q3 (Third Quartile)
=QUARTILE.INC(A2:A101,3)

Z-Score

=(A2-
AVERAGE(\$A\$2:\$A\$101))/STDEV.S(\$A\$2:\$A\$
101)

T-Distribution

=T.INV.2T(0.05,COUNT(A2:A31)-
1)*(STDEV.S(A2:A31)/SQRT(COUNT(A2:A31))
)

Probability Outside	Decimal	Used In
5%	0.05	95% confidence
1%	0.01	99% confidence
10%	0.10	90% confidence

Margin of Error

=1.96*(STDEV.S(A2:A51)/SQRT(COUNT(A2:A
51)))

Other Common Confidence Levels

Confidence Level	Z-value
90%	1.645
95%	1.96